A SYNOPSIS ON

MarketLens

Submitted in partial fulfilment of the requirement for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

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CANDIDATE'S DECLARATION

I/we hereby certify that the work which is being presented in the Synopsis entitled **"MarketLens"** in partial fulfillment of the requirements for the award of the Degree of **Master of Computer Applications** in the Department of Computer Applications of the Graphic Era (Deemed to be University), Dehradun shall be carried out by the undersigned under the supervision of **Dr. Bhawnesh Kumar**, Department of Computer Applications, Graphic Era (Deemed to be University), Dehradun.

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Chapter 1

Introduction and Problem Statement

1.1 Introduction

Social media is widely used by businesses to promote products, connect with customers, and understand current market trends. However, checking trends and competitor activities across different platforms can take a lot of time.

Market Lens is a web-based dashboard that brings this information together in one place. It helps users view marketing trends, track engagement, and compare competitor activities through a simple dashboard. The project uses **React, Java Spring Boot, MySQL, WebSockets, and Docker** to build an interactive and efficient system.

1.2 Problem Statement

Marketing information on social media is often spread across different platforms. Businesses have to manually check competitor posts, content, engagement, and posting patterns, which makes comparison difficult and time-consuming.

Market Lens aims to make this process easier by providing a single dashboard where users can organize, compare, and understand marketing data. DSA concepts will be used for efficient searching, sorting, and ranking of data, while real-time updates will keep the dashboard information current.

Chapter 2

Background/ Literature Survey

Social media has changed the way businesses connect with customers and promote their products. Platforms such as Instagram, Facebook, YouTube, and LinkedIn generate a large amount of information every day through posts, likes, comments, shares, and other activities. This information can help businesses understand what people are interested in and what type of content is getting more attention.

Today, businesses also keep an eye on their competitors to understand their marketing strategies. They may check what competitors are posting, how frequently they post, which topics they focus on, and how much engagement their content receives. Doing all of this manually can take a lot of time, especially when information has to be collected from different places.

Existing social media analytics tools provide features such as performance tracking, engagement analysis, and reports. However, these tools can have many features that may not be required for a simple competitor and trend analysis system. For a user who mainly wants to compare marketing activities and identify useful trends, a simpler dashboard can make the information easier to understand.

The idea behind **Market Lens** is to bring the important information together in one place. The dashboard will allow users to view marketing trends, compare competitors, and study engagement in a clear and organized way. Instead of going through large amounts of information manually, users will be able to understand the data through charts, comparisons, and structured results.

The project will also make use of **Data Structures and Algorithms** for handling the data efficiently. Searching and sorting techniques can be used to find specific information and arrange results, while ranking can help identify better-performing or trending content.

The system will be developed using **React** for the frontend, **Java Spring Boot** for backend operations, and **MySQL** for storing data. **WebSockets** will be used where real-time dashboard updates are required, and **Docker** will help run the application in a consistent environment.

Overall, Market Lens focuses on creating a practical and easy-to-use system where marketing information can be viewed and compared without making the process unnecessarily complicated.

Chapter 3

Objectives

The main objective of **Market Lens** is to develop a simple and useful dashboard that helps users understand social media marketing trends and competitor activities. The key objectives are:

1. **Competitor Analysis:**
   To compare competitors based on their content, posting frequency, topics, and engagement.

2. **Trend Tracking:**
   To identify and display popular or growing marketing trends in an easy-to-understand form.

3. **Data Management:**
   To organize marketing data using **DSA concepts such as searching, sorting, and ranking** for faster and better results.

4. **Interactive Dashboard:**
   To create a clean and user-friendly dashboard using **React** where users can easily view charts, comparisons, and important information.

5. **Real-Time Updates:**
   To use **WebSockets** for updating dashboard information in real time without repeatedly refreshing the page.

6. **Easy Deployment:**
   To use **Docker** so that the application can be set up and run easily on different systems.

Chapter 4

Expected Outcomes

The expected outcome of Market Lens is to create a simple and useful dashboard that makes social media marketing data easier to understand. The main outcomes of the project are:

1. Simple Dashboard:
   A clean dashboard where users can easily check important marketing information without going through too much data.

2. Easy Competitor Comparison:
   Users will be able to compare competitors based on their posts, content, posting activity, and engagement.

3. Better Understanding of Trends:
   The system will help users notice popular topics and understand what kind of content is getting more attention.

4. Quick Search and Sorting:
   DSA techniques will help users search, sort, and rank information so that useful results can be found quickly.

5. Live Dashboard Updates:
   Important information can be updated in real time using WebSockets, making the dashboard more interactive.

6. Easy to Run and Use:
   Docker will help the application run smoothly on different systems with less setup effort.

Overall, Market Lens is expected to save time and make it easier for users to understand and compare social media marketing activities from one place.

Chapter 5

Possible Approach/ Algorithms

• **Collecting the Data:**
The system will gather useful marketing information such as competitor posts, content topics, engagement, and posting activity.

• **Organizing the Information:**
The collected data will be stored in a structured way so that it is easy to manage and use whenever needed.

• **Search and Sort:**
Searching and sorting techniques will help users quickly find information and arrange it based on factors such as engagement or performance.

• **Comparison and Ranking:**
The system will compare competitors and their content to show which posts or activities are performing better.

• **Dashboard Updates:**
The results will be shown through simple charts and comparisons. WebSockets will help update the dashboard when new information is available.

• **Running the Application:**
Docker will be used to package the application so that it can be set up and run more easily on different systems.

References

[1] React Documentation – Used as a reference for building the user interface of the dashboard.
React Documentation

[2] Spring Boot Documentation – Referred to for developing and managing the backend of the application.
Spring Boot Documentation

[3] MySQL Documentation – Used for understanding database storage and management.
MySQL Documentation

[4] Java Documentation – Referred to for Java concepts used during backend development.
Java Documentation

[5] Docker Documentation – Used as a reference for containerizing and running the application.
Docker Documentation

[6] Spring WebSocket Documentation – Referred to for implementing real-time updates in the dashboard.
Spring WebSocket Documentation